

Koduru Suchith

Computer Science Engineering Graduate, Indian Institute of Technology Delhi '25

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PROFESSIONAL SUMMARY

AI & Full-Stack Engineer with **1+ year of experience** building production **LLM agents**, **RAG pipelines**, and **voice AI** at a fast-paced startup. Strong in **FastAPI**, **MCP servers**, and **real-time streaming** (WebSockets, SSE), with hands-on LLM orchestration (**AgentFlow**, **LangGraph**), vector search (**Qdrant**), and edge inference (**Jetson**, **Raspberry Pi 5**). Open-source contributor to **AgentFlow**. B.Tech, **IIT Delhi**.

EDUCATION

B.Tech, Computer Science & Engineering | Indian Institute of Technology Delhi 2021 – 2025

WORK EXPERIENCE

AI Software Engineer – 10xScale.ai | Hyderabad, India (June'25 – Present)

- Domain-Specialized Conversational Agent:**
- Built a domain-specialized agent on **AgentFlow** (LangGraph-like **StateGraph** + **ToolNode**), served via **FastAPI** with **SSE**-streamed responses to a **React** frontend.
 - Persisted state with a Postgres checkpointer for short-term context and **Qdrant** for long-term cross-session memory.
 - Authored custom transactional tools (scheduling, availability lookup, structured DB queries) so the agent takes real actions, not just chat.

- Voice-Enabled Edge AI Application (Raspberry Pi 5):**
- Built a voice-driven medical-kiosk assistant on **Raspberry Pi 5 (PySide6)** with an async **VAD** → **STT** → **Agent** → **TTS** pipeline and a **FastAPI** backend; supports 11 Indic languages auto-detected per turn.
 - Designed an **operator-in-the-loop** flow over **WebSockets** streaming live PCM audio to a **React** dashboard, with operator-first STT approval and timeout fallback.
 - Integrated **WhatsApp** via **Twilio**: inbound webhook extracts PDFs/images using **Gemini Vision** into agent context; outbound consultation summary via **Gemini 2.5 Flash**.
 - Reduced barge-in latency **~70%** (1000 ms → 300 ms) by flushing the OS audio buffer on stop-TTS and chunking TTS at the first sentence.

- MCP Server & Database-Aware Agent:**
- Built a custom **MCP server (FastMCP)** exposing **PostgreSQL** query and **Typesense** search tools for natural-language lookups over structured data.
 - Designed a database-aware reasoning agent that plans tool calls and grounds responses in real data, eliminating hallucinations on factual questions.

- AgentFlow – Open Source Contributions:**
- Contributed to open-source **AgentFlow**: shipped a **Qdrant**-backed long-term memory module and hardened the **Google GenAI** adapter against empty-response edge cases. **2 PRs merged** (#17, #19).

- General Experience:**
- Deployed services with **Docker** and configured routing, upstreams, and WebSocket passthrough in **Apache APISIX**.
 - Benchmarked local LLM inference on **NVIDIA Jetson Orin Nano** (Ollama 7B, MedGemma 4B, Whisper, Qwen TTS) for edge AI viability.

Maternal Health Risk Predictor | Summer Project, Prof. Kapil Tomar (May'24 – Jul'24)

- Benchmarked **4 classifiers** (SVM, Random Forest, XGBoost, Gradient Boosting); achieved **99% accuracy** with Gradient Boosting after hyperparameter tuning.
- Applied **SHAP** for model interpretability and **threshold optimisation** to align predictions with clinical decision objectives.
- Deployed a **Flask**-based web app with Predict, Result, and Dashboard pages; highlighted out-of-range inputs for enhanced clinical interpretability.

KEY PROJECTS

Multi-Class Image Classification (CIFAR-100) | Machine Learning, Prof. Rahul Garg (Sep'24 – Nov'24)

- Developed a high-confidence image classification model on the **CIFAR-100** dataset, experimenting with multiple architectures and calibration methods using **PyTorch**.
- Implemented **WideResNet (WRN-28-10)** and boosted generalisation using **Sharpness-Aware Minimization (SAM)**.
- Enhanced prediction reliability through **temperature scaling** calibration, improving calibrated accuracy on high-confidence outputs.

TECHNICAL SKILLS

Languages	Python, C++ , SQL, JavaScript/TypeScript	AI / ML	PyTorch, AgentFlow, LangGraph, LangChain, MCP / FastMCP, RAG, Gemini, SHAP, Scikit-Learn
Backend	FastAPI, async Python, PostgreSQL, Qdrant, Typesense, REST, WebSockets, SSE	Frontend	React, PySide6, Streamlit
DevOps	Docker, Git, GitHub Actions, Apache APISIX, GCP / GCS	Edge / LLMs	Jetson Orin Nano, Raspberry Pi 5, Ollama, Whisper, Sarvam STT/TTS, Twilio